Coffee, Exercise, and Reaction Time

A 2^2 Factorial Experiment with Blocking on Sex

STATS 101B · Introduction to Design and Analysis of Experiment · Spring 2026
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1. Introduction

How quickly can you catch a falling ruler — and could you do it faster after a cup of coffee or a quick burst of exercise? Reaction time is an everyday measure of alertness, and caffeine and physical activity are two of the most common things people reach for when they want to feel sharper. This project asks whether either actually helps.

Formally, our research question is whether drinking coffee (regular versus decaffeinated) and performing a short bout of exercise (30 seconds of jumping versus rest) significantly changes a person's reaction time, and whether the two act together. Going in, our expectation was that both caffeine and exercise would reduce reaction time, and that their interaction would matter as well — that being both caffeinated and physically primed would produce the fastest responses of all.

To answer this, we designed and ran a 2^2 factorial experiment with one blocking variable on the virtual population of The Islands. Each subject completed a ruler-drop reaction-time test before and after receiving an assigned combination of caffeine and exercise, and we analyzed the resulting percent change in reaction time using analysis of variance (ANOVA). The sections below describe the design and sampling plan, present the results, and discuss what they do and do not tell us about coffee, exercise, and reflexes.

2. Design of the Experiment

Design and factors. We used a 2^2 factorial design: two treatment factors, each at two levels, giving $2 \times 2 = 4$ treatment combinations. A factorial design is the natural choice here because it lets us estimate both main effects and the interaction between caffeine and exercise from a single experiment, rather than studying each factor in isolation. Following the convention from lecture, we code each factor's low level as −1 and its high level as +1.

Factor	Low level (−1)	High level (+1)
Caffeine (A)	Decaf, 250 mL	Regular coffee, 250 mL
Exercise (B)	Rest (no exercise)	30 seconds of jumping

The four treatment combinations are therefore (Decaf, Rest), (Coffee, Rest), (Decaf, Jump), and (Coffee, Jump).

Blocking. We blocked on sex (male / female). Sex is a fixed, pre-existing characteristic of each islander rather than something we manipulate, and it is a plausible source of variation in baseline physiology and reaction time. Treating it as a block, rather than ignoring it, removes that between-sex variation from the experimental error and lets us judge the treatment effects more cleanly. Because sex cannot be randomly assigned, it is a nuisance factor, not a treatment — exactly the situation blocking is designed for.

Response variable. Reaction time was measured with the ruler-drop test: a ruler is released and we record the distance d (in centimeters) at which the subject catches it. By the free-fall relationship, the corresponding reaction time is

$$t = \sqrt{(2d / g)}, \quad g = 9.8 \text{ m/s}^2$$

which we express in milliseconds. Because each islander has a different baseline speed, we do not analyze raw reaction time directly. Instead, our response variable is the percent change in reaction time from before to after treatment,

$$\% \text{ change} = (t_{after} - t_{before}) / t_{before} \times 100\%$$

where a negative value means the subject got faster. Working with proportional change removes each person's baseline reflex speed from the comparison, so we measure the effect of the treatment rather than who happened to be quick to begin with.

Sample and balance. Our design uses n = 4 × 18 = 72 subjects, with 18 in each of the four treatment combinations. Crucially, the design is balanced once the block is included: each of the eight Sex × Caffeine × Exercise cells contains exactly 9 subjects (8 × 9 = 72). Balance makes the design orthogonal, so that every main effect, the interaction, and the block are estimated independently of one another — adding or removing a term does not disturb the others. This matches the balanced factorial framework developed in lecture and makes the ANOVA straightforward to interpret.

Sampling procedure. Subjects were drawn from the town of Eden on the island of Bonne Santé, which has a residential population of N = 1463. We sampled as follows: (1) count the residents of each house and accumulate the counts to index the whole town; (2) draw a random integer from 1 to N without replacement and identify the house containing that resident; (3) sample only one person per house, re-drawing if a house already used comes up again; and (4) once data were collected, trim any Sex × Caffeine × Exercise cell that exceeded 9 subjects back down to 9, preserving the balanced design.

Sample size justification (power). We treated the 2^2 design as a one-way layout with k = 4 treatment groups and used the pwr package's pwr.anova.test to plan the sample size. Targeting 80% power at a significance level of α = 0.05 to detect a large effect (Cohen's f = 0.40 — a reasonable expectation for a clear behavioral manipulation), the calculation calls for roughly 19 subjects per group. Our balanced design provides 18 per group (72 total), which delivers power of approximately 0.80. In other words, trimming the sample to a perfectly balanced design costs only a negligible amount of power while buying us full orthogonality.

3. Results and Interpretation

The mean percent change in reaction time for each of the four treatment combinations is shown in Table 1. Two patterns stand out immediately: the two resting cells sit essentially at zero, while both jumping cells fall to roughly −13% to −14%.

	Rest	Jump 30 s
Decaf	−0.29%	−13.05%
Coffee	+0.07%	−13.92%

Table 1. Mean percent change in reaction time by treatment combination.

The side-by-side boxplots in Figure 1 tell the same story. For the two resting groups the median change hovers around 0%, whereas after 30 seconds of jumping reaction time drops by about 13–14% in both the decaf and coffee groups. The decaf-and-jump group contains a single outlier near 0%. The Decaf groups are fairly symmetric (aside from that outlier), while the Coffee & Rest group is slightly right-skewed and the Coffee & Jump group slightly left-skewed; within each exercise level the decaf and coffee medians sit close together even though their spreads differ.

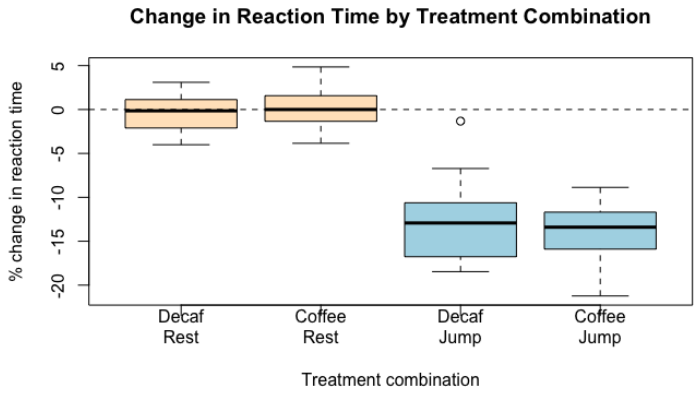


Figure 1. Percent change in reaction time by treatment combination. The dashed line marks 0% (no change).

ANOVA. We fit the model response ~ A * B + Block in R using aov(). The resulting analysis-of-variance table is given in Table 2.

Source	df	Sum Sq	Mean Sq	F value	p-value
Caffeine (A)	1	1	1	0.106	0.746
Exercise (B)	1	**3221**	**3221**	**306.033**	**< 2e-16 *****
Sex (Block)	1	1	1	0.051	0.821
A × B interaction	1	7	7	0.646	0.424
Residuals	67	705	11		

*Table 2. ANOVA for the 2^2 factorial design with sex as a block. Significance codes: *** $p < 0.001$.*

Only the Exercise factor (B) is statistically significant: its p-value is far below $\alpha = 0.05$ ($F = 306.0$, $p < 2 \times 10^{-16}$). Caffeine (A), the Caffeine × Exercise interaction (A × B), and the sex block are all clearly non-significant ($p = 0.746$, 0.424, and 0.821, respectively). The model explains about 82% of the variation in percent change in reaction time ($R^2 \approx 0.821$). Because the design is balanced, these conclusions are not sensitive to model specification — dropping the block leaves the other sums of squares unchanged, since every effect is estimated orthogonally.

Effect estimates. Refitting the model as a regression with lm() under the same ±1 coding, each estimated factorial effect equals twice the corresponding regression coefficient (effect = 2 × coefficient). The estimates are shown in Table 3.

Term	Effect (= 2 × coefficient)
Exercise (B)	**−13.378%**
Caffeine (A)	−0.248%
A × B interaction	−0.615%
Sex (block)	+0.174%

Table 3. Estimated factorial effects.

Retaining only the significant term, the fitted model reduces to

$$\hat{y} = -6.797 - 6.689 \cdot B$$

where the intercept is the grand mean of all recorded percent changes. The interpretation is direct: moving from rest to 30 seconds of jumping speeds reaction time by about 13.4 percentage points, while the caffeine

effect, the interaction, and the block are all under one percentage point in magnitude and statistically indistinguishable from zero.

The interaction and main-effect plots confirm this. In the interaction plot (Figure 2) the decaf and coffee lines are nearly parallel and very close together — they overlap only barely — indicating no meaningful Caffeine × Exercise interaction, and their steep common downward slope is the exercise effect. The main-effect plots (Figure 3) make the contrast explicit: the caffeine line is essentially flat, while the exercise line drops by roughly 13% from rest to jumping.

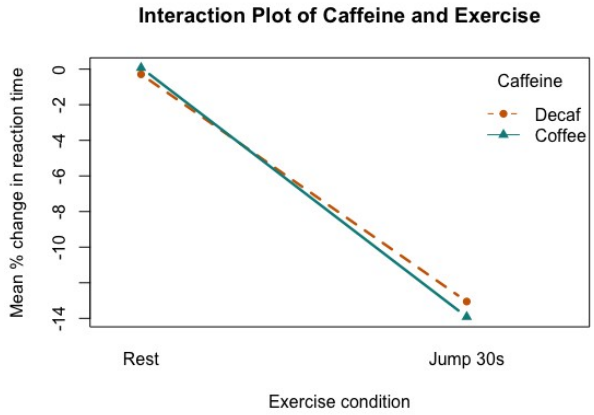


Figure 2. Interaction plot of caffeine and exercise; the two lines are nearly parallel.

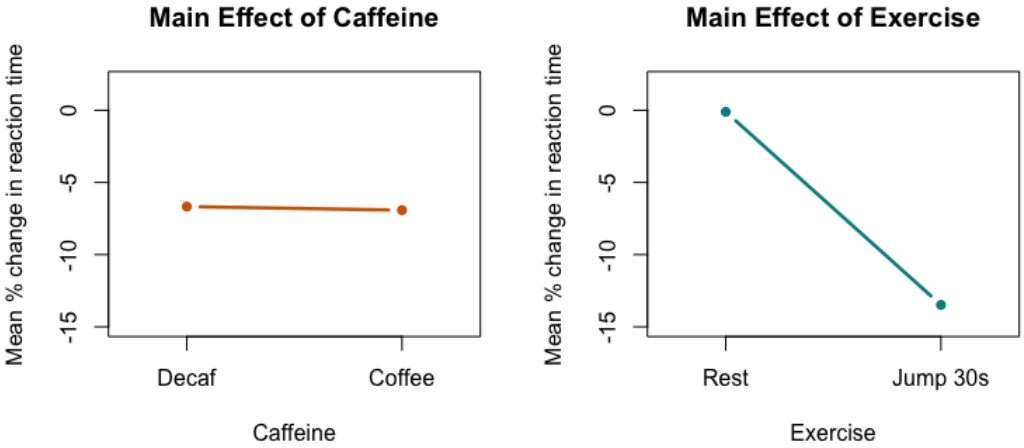


Figure 3. Main-effect plots for caffeine (left) and exercise (right).

Model diagnostics. Figure 4 checks the ANOVA assumptions. In the residuals-versus-fitted plot the residuals scatter evenly around zero with comparable spread at both fitted-value clusters, consistent with constant variance (homoskedasticity). In the normal Q–Q plot the residuals fall close to the reference line, so the normality assumption on the errors is reasonable. We are therefore comfortable relying on the ANOVA results above.

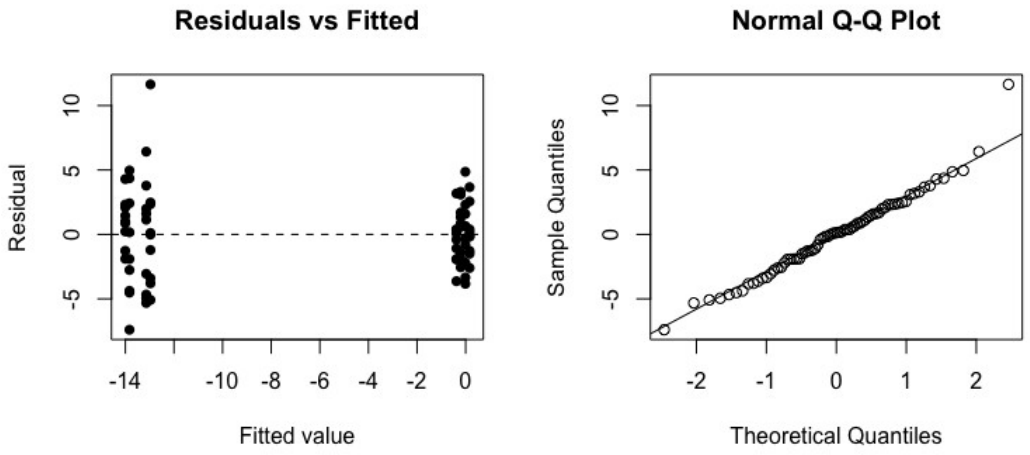


Figure 4. Model diagnostics: residuals vs. fitted (left) and normal Q–Q plot (right).

4. Discussion

Summary. A short bout of exercise has a large and highly significant effect on reaction time: 30 seconds of jumping made our subjects roughly 13.4% faster ($F \approx 306$, $p < 0.001$). Drinking a 250 mL regular coffee instead of decaf, by contrast, had no detectable effect ($F \approx 0.11$, $p \approx 0.75$); there was no interaction between the two ($F \approx 0.65$, $p \approx 0.42$); and the sex block was negligible ($F \approx 0.05$, $p \approx 0.82$). The practical takeaway is blunt: if your goal is to catch a falling ruler more quickly, the evidence says you are far better off jumping for 30 seconds than reaching for a cup of coffee.

Does it match the real world? These findings line up well with the literature. A short bout of moderate physical activity raises physiological arousal, and reviews of acute aerobic exercise report a reliable improvement in response time afterward, so a measurable speed-up after jumping is exactly what we would expect (Frontiers in Psychology, 2021). The null result for caffeine is also plausible once timing is considered: caffeine reaches peak plasma concentration only about 15–120 minutes after ingestion, with most absorption over the first 0–45 minutes (Institute of Medicine, 2001). Because we administered the second ruler test shortly after the subject drank, our measurement very likely preceded caffeine's peak effect — so finding no caffeine effect does not mean caffeine does nothing, only that it had not yet taken hold.

Limitations. Several features of the design limit how far the results generalize. First, the entire sample comes from a single town — Eden on Bonne Santé, which appears to have a tropical climate (thatched roofs, palm trees). Sampling additional towns on cooler islands such as Ironbard or Providence would reveal whether climate quietly influences reaction time and would broaden the scope of any conclusions. Second, the caffeine manipulation did not allow for digestion; a longer waiting period before the post-treatment test might have produced a larger, possibly significant, caffeine contrast. Third, our subjects ranged in age from 12 to 98, a very wide span, so age is a strong candidate for a second blocking variable (for example, grouping 0–50 versus 51–100), which would absorb additional variability and sharpen the treatment comparisons. Finally, the requirement of balance fixed the sample at a multiple of 8, and $n = 72$ was chosen to reach about 80% power for a large effect ($f = 0.40$); a design powered only for large effects can miss smaller but real ones, so a more subtle caffeine effect could have gone undetected here.

Future work. A natural second replication would (1) administer the post-treatment ruler test at least 15 minutes after the subject drinks the coffee, to capture caffeine nearer its peak; (2) sample several towns across islands of differing climates to improve external validity; and (3) add one or two further caffeine levels — for instance espresso, or espresso with sugar, standardized to 250 mL — turning caffeine into a multi-level factor and allowing a richer dose–response picture. Adding a second block on age and increasing replication would tighten the experiment further.

References

1. Institute of Medicine. *Caffeine for the Sustainment of Mental Task Performance: Formulations for Military Operations.* Washington, DC: National Academies Press; 2001.
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Code availability. The complete R Markdown code used to import the data, run the power calculation, fit the ANOVA and regression models, and produce all figures is submitted alongside this report as Stats_101B_Final_Project_Code.Rmd.